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Department of Electrical and Electronic Engineering

EEEE3001

Final Year Individual Project Proposal

Autonomous Robot for Assessing Wheelchair Accessibility

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1 Overview

Narrow doorways and obstacles restrict wheelchair access, yet building managers need measured evidence to prioritise improvements. VisitEngland estimates annual spending in England by visitors with health impairments and their companions at £14.6 billion, indicating the commercial value of accessible premises [1]. Where applicable, the robot will screen barriers against Approved Document M, England's access guidance for non-residential buildings [2], while leaving compliance decisions to qualified assessors.

Existing approaches include Room Accessibility and Safety Scanning in Augmented Reality (RASSAR), which overlays accessibility findings on smartphone views [3], and the Robotic Accessibility Indoor Scanner (RAIS), which automates surveys using a vacuum robot and smartphone [4]. This project combines calibrated stereo vision, which estimates depth from two cameras, with ultra-wideband (UWB) positioning. Fixed UWB anchors locate a tag carried by the robot, linking measured doorway widths and obstacles to an auditable map. Existing laboratory UWB equipment and affordable cameras help limit hardware costs; the combination will be tested for repeatable measurements across surveys.

Aim. Develop and evaluate a compact robot that autonomously surveys controlled indoor routes, measures potential wheelchair-access barriers, and presents an annotated map and inspection report through a live web interface. The laboratory provides a Raspberry Pi 5 computer and UWB equipment. The selected Waveshare UGV01 tracked vehicle provides encoders and an ESP32 microcontroller for motor control [5].

Robot Operating System 2 (ROS 2) connects sensing, positioning and reporting software on the Ubuntu Linux operating system. Figure 1 shows these dependencies: dimensions require calibrated cameras, and mapping also needs robot position and heading. The ESP32 controls motor speed and stops the motors if Pi commands cease. This independent stop protects against loss of communication or stalled software.

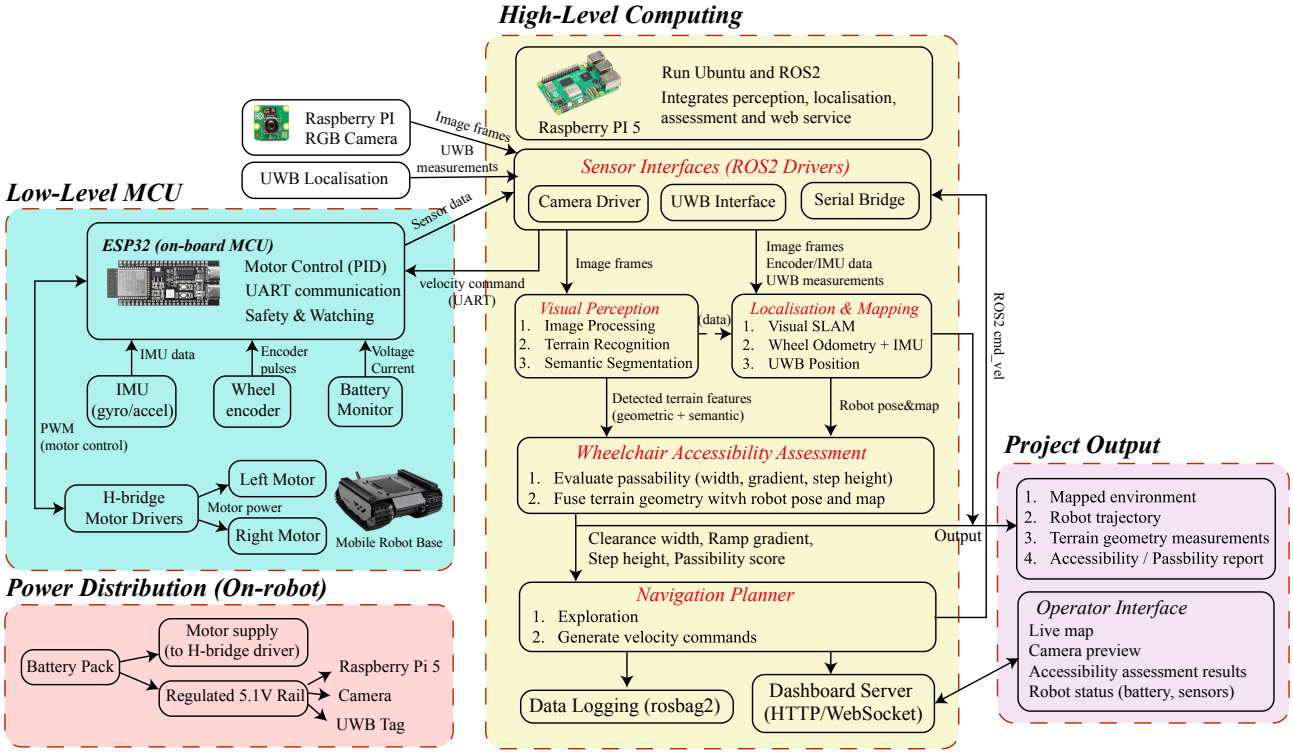


Figure 1: Project system diagram

Tests will use dry indoor routes with stationary obstacles within a surveyed UWB zone. In Figure 1, the cameras identify features and measure clearances; wheel motion, an inertial measurement unit (IMU) and UWB locate findings on the map. The planner uses obstacles and robot position to choose a route; ROS 2 logs evidence for a web dashboard. Doorway clearance and obstacle records are essential; ramp and threshold measures are stretch goals. The diagram's passibility score is conceptual, not an essential deliverable or an individual access verdict.

2 Specification

Essential requirements take priority over stretch work. Difficulty labels identify the main challenge. Unless sourced, numerical limits are provisional engineering targets to be checked in pilot trials, not established

performance. Use 200–800 lx (lux, illuminance) as a controlled indoor test envelope and capture dimensions while stationary. Repeated trials assess consistency within this envelope, not population-wide reliability.

2.1 Essential: hardware and safe operation

- **Mobility (moderate: mounting/stability).** The $\leq 300 \times 250$ mm footprint (plan-view size) allows mounting around the selected 231×197 mm chassis. Limit added equipment to 1.0 kg, conservatively below the advertised 6.2 kg carrying capacity [5]; weigh and measure the assembly. Pass ten ascents and ten descents of an 8° ramp, and ten 15 mm step climbs without assistance, tipping or grounding. These reduced targets allow for mounting effects relative to the advertised 36°/40 mm capability.
- **Stopping (high: fault handling).** At ≤ 0.25 m/s, disable drive within 0.20 s of the last valid command and stop within 0.20 m: 50 mm timeout travel plus a provisional 150 mm braking allowance. Pass ten tests each of emergency stop, communication loss and stale perception (outdated sensor data), measuring trigger-to-rest distance and time; require manual re-arm.
- **Power (moderate: voltage dips).** Use a regulated 5.1 V, 5 A-capable Pi supply following manufacturer guidance [6]. Pass three 20-minute loaded runs with cameras, motors and processing active, without undervoltage or resets; this tests sustained operation over repeated short surveys.

2.2 Essential: measurement and autonomous survey

- **Clearance (high: calibration/edges).** Doorway mean absolute error (average unsigned deviation) ≤ 30 mm; 95th-percentile error ≤ 60 mm (95% of errors within this value); valid readings in $\geq 90\%$ of attempts. The stereo estimate in Section 3 motivates centimetre-scale screening, with a provisional ± 60 mm human-review band. Test ten 0.65–1.20 m openings spanning restricted and wider access, at 1.5 m and $0, \pm 5°$ approach angles, three repeats: 90 attempts against a reference accurate to 2 mm.
- **Recognition (high: unseen scenes/speed).** For both doorways and obstacles, precision (correct/all detections) and recall (detected/all actual features) must reach 0.85: a screening target limiting both error types. Test at least 50 unseen instances per class, giving 2-percentage-point recall increments. Match detections one-to-one to labelled objects of the same class, requiring overlap area divided by the area covered by either box ≥ 0.5. Stereo and detection must each run at ≥ 5 Hz (updates per second), with 95th-percentile image-to-result delay ≤ 200 ms: 50 mm nominal travel per update. Check Pi timestamps; obstacle stopping uses depth independently of labels.
- **Location (high: reflections/slip).** Position root-mean-square error (square root of average squared error) ≤ 0.20 m and mean absolute heading error $\leq 5°$ target finding-level map accuracy. At 1.5 m, 5° adds about 0.13 m lateral error; map locations therefore cannot establish doorway clearance. Test 20 surveyed positions/headings, each visited three times across a 5×5 m laboratory anchor zone.
- **Routes (high: integration).** Complete a 10 m route with three obstacles in nine of ten trials without intervention, and with no collision or edge crossing in any trial. This compact course exercises repeated avoidance and turning. Finish within 0.30 m of the surveyed goal, an independently tested tolerance allowing control error beyond the positioning target. Log paths, stops and interventions.
- **Report (moderate: alignment).** Refresh the map at ≥ 1 Hz, limiting nominal display displacement to 0.25 m per update; stopping uses the faster sensor stream. All 20 audited findings must match logs for location, dimensions, uncertainty, time and rule source, testing traceability across the surveyed zone.

2.3 Stretch: added assessment capability

First: ramps/thresholds (high: floor estimation). Target mean absolute slope/height errors $\leq 1°/\leq 5$ mm for screening small changes. Test five heights over 0–30 mm and four slopes over 0–10°, ten readings each at 0.4–0.7 m, against a caliper and angle meter. Short range improves stereo depth sensitivity; climbing ability is not a wheelchair-access criterion. **Then: fewer false flags (high: demonstrating benefit).** Seek $\geq 20\%$ relative reduction in false barrier flags versus geometry alone, without reduced barrier recall or valid-data coverage, on ≥ 100 paired cases reserved for testing. This substantial improvement target justifies added recognition complexity; report counts and uncertainty if few false flags occur.

3 Methodology

Develop safe control, doorway measurement, positioning and reporting in stages. Bench tests and recorded-data validation precede supervised driving; reserve 32 hours for calibration, model training and sensor-fusion exercises.

3.1 Platform and software interfaces

Complete laboratory induction and risk assessment before powered trials. Mount equipment on the UGV01 rails; verify loaded stability, motor-start power and independent shutdown. Fit a fused battery, guards and a latching emergency stop requiring manual reset; use a suitable charger. Test restrained ramps and a guarded mock edge away from public routes.

ROS 2 provides time-stamped messages, recorded by its rosbag2 tool for replay; a serial bridge links Pi and ESP32. Prove Ubuntu dual-camera capture and command-timeout stopping early, as driver support and timing constrain later stages. The ESP32 stops if Pi commands cease; the Pi withdraws motion permission when perception is outdated. Web updates remain outside the stopping loop.

3.2 Stereo measurement and feature recognition

Lock focus on two Camera Module 3 units 120 mm apart [6], balancing depth sensitivity with chassis width. Open Source Computer Vision (OpenCV), an image-processing library, calibrates stereo geometry from 25 checkerboard views; target mean reprojection error (observed–predicted image-point distance) below 0.5 pixel. Semi-global block matching recovers depth from image patches [7]. For each doorway, locate the visible opening edges, calculate width from calibrated camera rays and their depths, and reject occluded or uncertain edges. At 1.5 m, a provisional 500-pixel focal length and half-pixel matching error imply about 19 mm depth error, excluding edge and calibration errors. Verify camera parameters and width errors. Reserve 8 h for calibration practice; stationary capture limits timing errors, while blank or poorly lit surfaces require rejected depth.

Adapt a Single Shot MultiBox Detector (SSD) with MobileNetV2, a compact image-feature network, using pretrained TensorFlow machine-learning models [8]. This artificial intelligence (AI) stage labels doorways and obstacles. Start with about 300 labelled images; split distinct scenes 60%/20%/20% for training/tuning/testing to prevent inflated accuracy. Expand collection until 50 test instances per class are available. Reserve 12 h for labelling/training exercises. Train on a workstation; benchmark accuracy and speed on the Pi.

3.3 Localisation and navigation

Survey three well-spaced UWB anchors, the minimum for planar ranging with known tag height; installed anchors limit deployment to the test zone. An extended Kalman filter, which combines uncertain measurements in a changing motion model, merges encoder travel and IMU turning with UWB drift corrections. Check reflections and slip against surveyed references before driving. Reserve 12 h for filter exercises; camera-based motion remains optional.

A*, a grid path-search algorithm, plans to an operator-selected goal using 50 mm cells, matching nominal travel per sensor update. Mark cells clear, occupied or unknown and avoid unknown space. Enlarge obstacles by robot size, position uncertainty and stopping clearance; a 200 ms sensor delay adds 50 mm to the 0.20 m stop allowance before uncertainty margins. Stop on older frames or unreliable position. Use a pre-surveyed room boundary and update obstacles from stereo; recorded-data tests must pass before live routes.

3.4 Reporting and evaluation

A Python-based web dashboard will display doorway clearances, obstacles, robot positions and operating status. Images, timestamps, positioning data and calibration/model versions will be recorded to support validation and reproducibility. Where applicable, findings will be compared against relevant accessibility guidance [2]. The applicable doorway rule depends on door and approach context; if this context is missing or the uncertainty band overlaps the rule threshold, report dimensions and flag the finding for manual review. Ramp and threshold measurements remain stretch goals, and individual wheelchair requirements remain outside the project scope.

Evaluate Section 2 targets, recording measurement errors, recognition results, failed trials and test conditions. Deliver the tested robot, source code, calibration data, measurement maps, reports and a dissertation recording achieved and unmet requirements. Avoid identifiable individuals during data collection; follow university storage requirements and record software and model licences.

4 Risk Management and Mitigation

Rank by relative score = initial chance (use 50 for 50%) × impact (1–5): 5 unsafe stop, 4 blocked build/tests, 3 missed core result, 2 recoverable evidence loss, 1 recoverable delay. Estimate likelihood as low (20%, routine), medium (30–40%, unresolved checks), or high (50–60%, untested critical work). Each title shows its score, e.g. 50 × 4 = 200. In each **Risk** line, the period gives when it may occur; 50%→20% gives the chance of the stated delay before→after verified mitigation; "up to 2 weeks" is possible time lost. Review estimates at supervisor reviews R1–R5.

- **Parts unavailable** (50 × 4 = 200)
  Risk: Oct–Dec; 50%→20%; up to 2 weeks. A missing chassis or camera blocks assembly or calibration. **Mitigation:** Check stock/cost; use replaceable parts and a spare camera if affordable. If not shipped by 15 November, borrow an alternative within budget; process images on the bench meanwhile.
- **Loaded robot fails mobility or power checks** (50 × 4 = 200)
  Risk: Nov–Dec; 50%→20%; up to 1 week. Extra mass, loose mounts or voltage drops when motors start can block commissioning. **Mitigation:** Weigh and test stability/voltage under load before ramp trials; reposition equipment, reduce mass or improve the power supply, then retest.
- **Stereo measurements are unreliable** (60 × 3 = 180)
  Risk: Dec–Feb; 60%→20%; up to 2 weeks. Camera movement, unsynchronised images or blank surfaces can distort doorway widths. **Mitigation:** Secure and calibrate the cameras, measure test doorways of known width while stationary and reject uncertain depth. If the 26 January measurement check fails, fix it before adding stretch features.
- **Stopping test fails** (30 × 5 = 150)
  Risk: Dec; 30%→10%; up to 1 week. A failed motor cut-off could leave the robot moving after communication is lost, so live trials must pause. **Mitigation:** Bench-test command loss and emergency stopping, then repeat with the loaded robot and check that restarting needs a manual reset. Use recordings during repairs; drive only after the stop tests pass.
- **Training data is inadequate** (50 × 3 = 150)
  Risk: Dec–Feb; 50%→20%; up to 2 weeks. Too few varied scenes or inconsistent labels can cause missed barriers. **Mitigation:** Collect separate training and test scenes, review labels and keep 50 test examples per class out of training. Retest missed cases; if recognition still fails, report measured geometry and mark recognition incomplete.
- **Navigation does not integrate** (50 × 3 = 150)
  Risk: Feb–Apr; 50%→20%; up to 2 weeks. UWB position drift misplaces obstacles; slow Pi processing delays reactions. Either can prevent autonomous routes. **Mitigation:** Replay journeys and measure processing time before driving. Move anchors or reduce model load, then retest in a re-surveyed area. If the 17 April check fails, demonstrate a supervised survey and report unmet autonomy.
- **Test space is unavailable** (30 × 4 = 120)
  Risk: Mar–Apr; 30%→10%; up to 1 week. Lost room access blocks planned doorway and route tests until another room is arranged. **Mitigation:** Book trials early and confirm access at R4. Prepare portable test doorways and an approved alternative room; re-survey anchor positions after moving and report any reduced test coverage.
- **Report links the wrong evidence or rule** (40 × 3 = 120)
  Risk: Mar–Apr; 40%→20%; up to 1 week. A wrong time, position or doorway rule can make an accurate measurement misleading. **Mitigation:** Replay logs and check 20 report findings against images, positions and rule sources. Flag missing context for human review; if records disagree, show measurements and uncertainty without an unsupported judgement.
- **Laptop or data is lost** (20 × 2 = 40)
  Risk: Throughout; 20%→10%; up to 1 week. Laptop failure or corrupted logs can erase assessment evidence. **Mitigation:** Version code; back up images, calibration and logs daily to university cloud storage and a local drive. Test restoration weekly; use a university computer if needed.
- **Personal time is lost** (30 × 1 = 30)
  Risk: Throughout; 30%→20%; 1–2 weeks. Illness, exams or disrupted travel reduce time for integration, retesting and writing. **Mitigation:** Write throughout and keep files accessible remotely. If a week is lost, tell the supervisor and replan using the shared 24 h reserve; discuss scope if disruption persists.

5 Time Plan

Figure 2 links platform, sensing and positioning to Section 2 targets. Each bar includes development, testing and documentation. Live routes follow validated control and sensing; fixture, replay and dashboard work can continue during hardware delays.

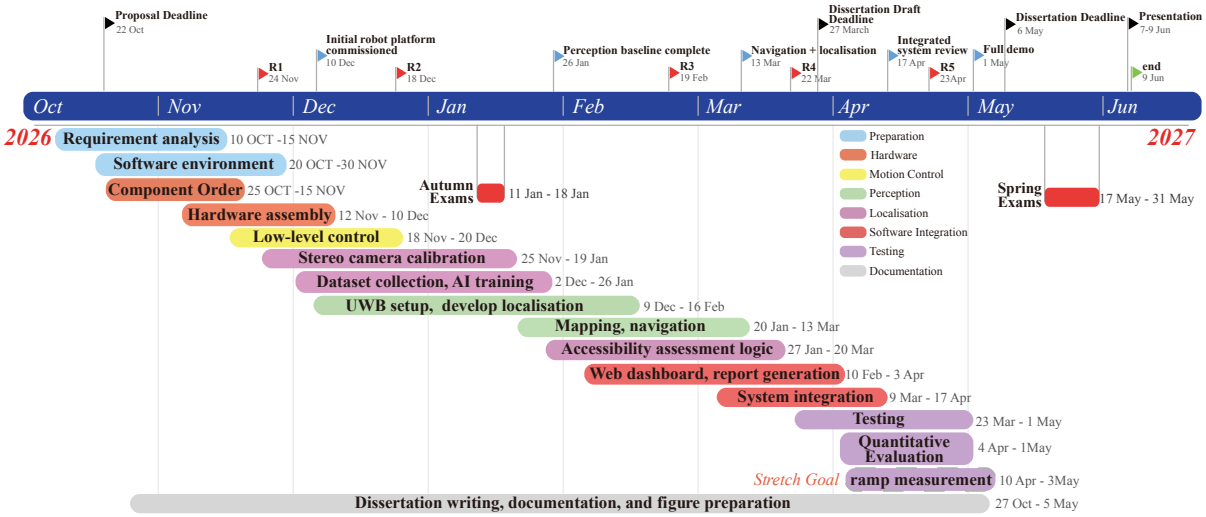


Figure 2: Project schedule; R1–R5 mark supervisor reviews.

Reviews. Agree dates with the supervisor; record evidence, risks and actions.

- **R1, 24 Nov:** Requirements/components checked; resolve delivery, camera and power faults before 10 Dec.
- **R2, 18 Dec:** Loaded stop/power tested; remote driving nearly complete; review stereo and UWB faults.
- **R3, 19 Feb:** Perception/positioning baselines tested; fix accuracy and timing before live routes.
- **R4, 22 Mar:** Pilot routes/report ready; confirm safety, evidence links and room access before 27 Mar draft.
- **R5, 23 Apr:** Integrated report and evaluation underway; close 17 Apr gaps before 1 May demonstration.

Workload. Budget 280 h: 188 technical (40 platform, 56 vision, 36 positioning, 32 navigation, 24 reporting; including 32 learning), 44 documentation, 16 presentation, 8 stretch and 24 contingency. This averages 8.5 h/week over 33 weeks. Shared weekly caps are 12 h in teaching, 4 at Christmas, 8 at Easter and 2 during assessments. These are initial capacity estimates: review actual hours weekly, sharing capacity across overlapping bars. Finish writing 5 May, submit 6 May and present 7–9 June.

Uncertainty. Figure 2 shows target dates, not guaranteed finishes. Delays in hardware, sensing, positioning or integration can move dependent tasks; update the schedule at each supervisor review using measured progress. Parallel fixture, replay and dashboard work limits idle time. The shared 24 h reserve covers at most two teaching weeks of extra work across the project, not two weeks for each task. Protect the final week before 1 May for repeat tests. Drop stretch work before using the reserve; agree reduced scope if essentials still overrun.

6 References

[1] VisitBritain/VisitEngland, *Accessible and inclusive tourism*, "The value of the Purple Pound (2023/2024)."
[2] Ministry of Housing, Communities and Local Government, *Approved Document M, Vol. 2*, 2015 ed., 2024 amendments, Tables 1–2.
[3] X. Su et al., "RASSAR: Room Accessibility and Safety Scanning in Augmented Reality," *Conf. on Human Factors in Computing Systems*, 2024.
[4] X. Su, D. Campos Zamora and J. E. Froehlich, "RAIS: Towards A Robotic Mapping and Assessment Tool for Indoor Accessibility Using Commodity Hardware," *Conf. on Computers and Accessibility*, 2024.
[5] Waveshare, *UGV01 product specifications* and *UGV01 Wiki* (tracks, payload, encoders and power).
[6] Raspberry Pi Ltd, *Computer hardware* and *Camera documentation* (Pi 5 power and Camera Module 3).
[7] OpenCV, *Semi-global block matching* and *Depth Map from Stereo Images*, 4.x.
[8] TensorFlow, *TensorFlow 2 Detection Model Zoo* (pretrained SSD MobileNet V2).

Statement of originality

- I have read and understood the University of Nottingham Academic Misconduct Policy.
- I am aware that failure to act in accordance with the Regulations Governing Academic Integrity may lead to the imposition of penalties which, for the most serious cases, may include termination of programme.
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You must change the statements in the boxes to reflect your individual submission.

You must acknowledge all sources of information (e.g. ideas, algorithms, data) using citations to auditable, authoritative sources.
You must also put quotation marks around any sections of text that you have copied without paraphrasing and cite the original source.
If any figures or tables have been taken or modified from another source, you must explain this in the caption, and cite the original source.

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If you have used any code (e,g. open-source code), reference designs, or similar resources that have been produced by anyone else, you must list them in the box below. In the report, you must explain what was used and how it relates to the work you have done.

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If you have used generative AI tools in the preparation of your assessed work, or the underlying code/designs/underpinning work, its use must be declared in a signed statement in the box below. You must declare the tools employed and the method and extent of use. You can do this You can do this using the table below:

Generative AI tools have been used in the preparation of this work as follows:

Tools	*Method*	*Extent of use*
Chat GPT	\	\
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.....		

I have acknowledged, in context, the use of all generative AI tools in preparation of this assessment, including textual, referencing, hardware, software, and intellectual inputs to my submitted work. All outputs were critically reviewed, edited, and verified by me.

You can consult with teaching staff/demonstrators, but you should not show anyone else your work (this includes uploading your work to publicly accessible repositories e,g. Github, unless expressly permitted by your supervisor). Nor should you help other students with their work.
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You must not have fabricated, modified or distorted any data, evidence, references, experimental results, or other material used or presented in the report. You must clearly describe your experiments and how the results were obtained, and include all data, source code and/or designs (submitted as a separate file) so that your results could be reproduced.

The material in the report is genuine, and I have included all my data/code/designs.

If your work involved research studies (including surveys) on human participants, their cells or data, or on animals, you must have been granted ethical approval before the work was carried out, and any experiments must have followed these requirements. You must give details of this in the report, and list the ethical approval reference number(s) in the box below.

Ethics approvals here. (Or indicate “not applicable”)

My responses to these questions are full and complete, and I have discussed them with my supervisor.	Student signature	Enze TU
	Date	Oct 19 2026